

Database Management System (CAC255)

Course Description

This course offers both theoretical as well as practical knowledge of database management system so that students can handle back end of software while developing any types of application packages.

Course Objectives

The general objectives of this subject are to provide the basic concept, theory and practices in design and implementation of DBMS. Students will also be good for handling different type of data transaction by using SQL commands.

Course Contents

Unit 1: Introduction to DBMS (3 Hrs.)

Introduction of Database Management System, Objective of Database Management System
Importance of DBMS, Merit and Demerit of DBMS, Application of DBMS

Unit 2: Database Design, Architecture and Model (6 Hrs.)

Overview of The Database Designing Process and View of Data

Structure of Database Management System, Level Database Architecture and Data Independence

Database Languages: DDL, DML, QBE

Data Models: Hierarchical, Network, Relational, E-R Model, Object Base Data Model

E-R Diagram: Concepts, Relationship, Entity Relationship Diagram, Weak Entity Sets, Strong Entity Set, Aggregation, Generalization, Converting ER Diagrams to Tables

Unit 3: Relational Database Model (4 Hrs.)

Structure of RDBMS and Terminology

Database Schema and Schema Diagram

Keys: Super, Candidate, Primary, Foreign, Composite etc. and Relationship

Introduction to Relational Algebra

Relational Algebra Operations: Select, Project, Cartesian Product, Union, Set Difference, Natural Join, Outer Join

Unit 4: Database Normalization (4 Hrs.)

Definition and Importance of Normalization, Functional Dependencies, **Normalization:** 1NF, 2NF, 3NF, BCNF and 4NF

Unit 5: Creating and Altering Database and Tables (SQL) (6 Hrs.)

Introduction to SQL, Creating Database with Different Type of Arguments and Alter Database

Creating Normal Tables and Complex Tables with Different Type of Constraints (Key, Check, Default)

Alter Tables: Adding and Dropping Attributes and Other Constraints

Drop Statement: Table, Database

Unit 6: Manipulating and Querying Data (8 Hrs.)

Adding Data with INSERT Statement

Retrieving Data with SELECT Statement and FROM Clause and Filter Data with WHERE Clause

Order and Grouping Data with ORDER and GROUP by Clause and Summarizing the Select Statement

Retrieving Data from Different Tables using: INNER JOINS, OUTER JOIN and CROSS JOIN

Building Nested Queries

Manipulate Data Using UPDATE Statement and Removing Rows Using DELETE Statement

Creating and Altering View

Unit 7: Developing Stored Procedures, DML Triggers and Indexing (5 Hrs.)

Managing Stored Procedures: Create, Alter, Drop, Execute Stored Procedure, Encryption, Passing Data To Stored Procedures, Parameter Default, Returning Data From Stored Procedure Output Parameter Using The Return Statement

Transaction Flow

Triggers: Creating Triggers, Triggers Limitation, Disabling Trigger, Developing Multi Row Enabled Triggers

Indexing: Basic Concept of Indexing, Ordered Indices, Type of Indexing, Multiple Key Access, Creating And Dropping Index

Unit 8: Query Processing and Security (5 Hrs.)

Overview of Query Processing, Measuring of Query Cost

Selection Operation, Sorting, Joining

Evaluation of Expression, Query Optimization

Database Administrator: DBA Roles and Responsibilities

Database Security Issues, Types of Security

Access Protection, User Accounts and Database Audits

Discretionary Access Control, Mandatory Access Control

Data Encryption And Decryption

Unit 9: Transaction and Concurrency Control (4 Hrs.)

Transaction Concept, Simple Transaction Model

Atomicity and Durability, Transaction Isolation

Serializability, Transaction Isolation and Atomicity and Transaction Isolation Levels

Lock-Based Protocol, Dead-Lock Handling, Multiple Granularity

Time-Based Protocol

Laboratory Works

Lab works should be done covering all the topics listed above using Oracle and a small project work should be carried out using the concept learnt in this course. Project should be assigned on Individual Basis.